a **Methodological Ground Simulation Standards (3-Chamber Parity)**

Chamber 1: Geomagnetic Baseline (GMF)

Magnetic Regime: 45–50 μT Ambient Terrestrial Field

Experimental Purpose: Terrestrial environmental control (calibrated local vector)

Chamber 2: Near-Null Field Simulation (NNMF)

Magnetic Regime: <10 nT (Shielded / Actively Cancelled)

Experimental Purpose: Lunar surface magnetic simulation (<0.8 nT ambient void)

Chamber 3: Matched Bifilar Sham Control

Magnetic Regime: Opposing windings; net B = 0

Experimental Purpose: Controls for ohmic heat, acoustic vibration & power-supply ripple

b **Hardware Self-Magnetic Assessment Protocol (Power-State Mapping)**

Subsystem Power-State Magnetic Characterization & Deconvolution

Subsystem 1: LED PAR Lighting Array (0–400 μmol/m²/s)

Power: 0 to 25 W (DC driver current 0.1–1.8 A)

Induced Field: ΔB_LED ≈ 2–15 μT; proportional to photon flux; balanced routing minimizes loop area

Subsystem 2: Fluidic & Nutrient Delivery Pumps / Valves

Power: 0 to 8 W (peristaltic/piezo motor pulses, 0.5–1.5 A peak)

Induced Field: ΔB_pump ≈ 8–35 μT; transient DC spikes during scheduled root module irrigation cycles

Subsystem 3: Thermal Control (Heaters & Thermoelectric Coolers)

Power: 0 to 30 W (resistive strips / TEC elements, 1.0–2.5 A continuous)

Induced Field: ΔB_thermal ≈ 12–48 μT; dominant DC contributor during cold lunar nights; bifilar loops required

Subsystem 4: Avionics, Telemetry & Sensor Power Bus

Power: 3 to 12 W (quiescent flight computer + RF comm transmission)

Induced Field: ΔB_bus ≈ 0.5–4 μT; static baseline shift + RF ripple; mapped via quiescent baseline calibration

$$\mathbf{B}_{ambient}(t) = \mathbf{B}_{sensor}(t) - \sum_{k} \Delta\mathbf{B}_k(P_k(t))$$ **: Computational Self-Field Subtraction**

c **Post-Flight Phenotypic Risk Disambiguation Framework**

Operational Multi-Stressor Disambiguation Matrix for Space Plant Biology

Near-Null Magnetic Field (<0.8 nT)	Lunar Partial Gravity (0.166 g)	Deep-Space Radiation (GCR / SPE)
Primary Biophysical Target:	**Primary Biophysical Target:**	**Primary Biophysical Target:**
Cryptochrome flavin radical pairs (FAD/TrpH•+); mitochondrial [4Fe-4S] cluster electron transfer.	Columella statolith sedimentation kinetics; mechanosensitive ion channels (MSL); PIN polar auxin.	Nuclear & organellar DNA double-strand breaks; radiolytic hydroxyl free radical (OH•) overflow.
Diagnostic Phenotypic Markers:	**Diagnostic Phenotypic Markers:**	**Diagnostic Phenotypic Markers:**
• Emergence lag (Δt50 = +12 to 36 h); capacity -10 to -30% • Repressed chlorophyll (HEMA1, POR) & anthocyanin (CHS) • Mitochondrial ROS leak; induction of AOX1D & CAT3 • Normal nuclear DNA integrity (zero γ-H2AX foci)	• Germination capacity & greening rate are normal • Directional root skewing, waving, & loss of g-angle • Cell wall remodeling (pectin demethylesterification) • Normal CRY signaling; localized apoplastic ROS (RBOH)	• Severe DNA double-strand breaks & γ-H2AX foci • ATM/ATR activation; chromosome aberrations & G2/M arrest • Widespread systemic oxidative stress & lipid peroxidation • Emergence fails only at acute lethal doses (>50 Gy)
Disambiguation & Rescue Strategy:	**Disambiguation & Rescue Strategy:**	**Disambiguation & Rescue Strategy:**
• Correlate with sample-level 3-axis canopy magnetometry • Fully rescued by 45 μT vector re-application • Completely insensitive to 1g centrifuge spinning	• Fully rescued to Earth phenotype by 1g centrifuge • Differential 1g vs μg controls in flight centrifuge • Completely unaffected by magnetic vector modulation	• Independent of magnetic field or gravity vector • Strictly correlates with canopy radiation dosimeters • Mitigated by shielding & chemical radioprotectants
[Diagnostic: Rescued by 45 μT | Insensitive to 1g]	**[Diagnostic: Rescued by 1g | Insensitive to B-field]**	**[Diagnostic: Correlates with Dosimeter | DNA Breaks]**